

Dataset : <https://docs.google.com/spreadsheets/d/1LXuheMo6rwzA1yq1r-gN5sKqTFcWZG7J/edit?usp=sharing&oid=102936154345016253077&rtpof=true&sd=true>

KPMG Data Analysis using Excel – Exact Project Questions

Task 1: Data Cleaning (12 Marks)

1. Customer Address Data

Remove any duplicate records.

Ensure all state names are correctly formatted.

2. Customer Demographic Data

Identify and correct any erroneous data entries (e.g., invalid characters in default).

Standardize the format for missing data entries.

Correct any anomalies in gender representation.

3. Transaction Data

Ensure that transaction_date is in a consistent date format.

Remove any records with missing or incomplete information.

4. New Customer Data

Standardize address formatting.

Ensure consistent gender representation.

Correct any anomalies in job_title and job_industry_category.

Task 2: Customer Segmentation (12 Marks)

1. Segmentation by Wealth Segment

Show the number of customers in each wealth_segment.

Calculate the average tenure for each wealth_segment.

2. Segmentation by Gender

Show the number of customers by gender.

Calculate the average past_3_years_bike_related_purchases for each gender.

3. Segmentation by Job Industry

Show the number of customers in each job_industry_category.

Analyze the distribution of wealth_segment within each industry.

Task 3: Transaction Analysis (12 Marks)

1. Sales Trend Analysis

Create a chart showing the total sales per month.

Identify any seasonal trends or significant spikes in sales.

2. Product Performance Analysis

Show the total sales for each brand.

Calculate the total sales and average list_price for each product_line.

3. Customer Purchase Behavior

Identify the top 10 customers based on total transaction value.

Calculate the average number of purchases per customer.

Task 4: New Customer Insights (12 Marks)

1. New Customer Demographics

Show the distribution of new customers by `wealth_segment` and `job_industry_category`.

Calculate the average `past_3_years_bike_related_purchases` for new customers.

2. New Customer Location Analysis

Create a map or chart showing the distribution of new customers by state.

Analyze the correlation between `property_valuation` and customer `wealth_segment`.

3. Potential Revenue from New Customers

Estimate potential revenue based on `past_3_years_bike_related_purchases` and value.

Task 5: Customer Lifetime Value (CLV) Analysis (16 Marks)

1. CLV Calculation

Calculate Customer Lifetime Value (CLV) for each customer using the formula:

$CLV = (\text{Average Purchase Value} \times \text{Purchase Frequency}) \times \text{Customer Lifespan}$

Where:

$\text{Average Purchase Value (APV)} = \text{Total Revenue} / \text{Number of Purchases}$

$\text{Purchase Frequency (PF)} = \text{Total Number of Transactions} / \text{Number of Unique Customers}$

$\text{Customer Lifespan (CL)} = \text{Tenure}$

2. Segment CLV Analysis

Show average CLV by `wealth_segment`.

Analyze the relationship between CLV and customer demographics (e.g., gender, job industry).

Task 6: Executive Summary and Recommendations (16 Marks)

Summarize the key findings from customer segmentation, transaction analysis, new customer insights, and CLV analysis.

Provide recommendations for marketing strategies targeting high-value customer segments.

Suggest potential areas for business expansion based on new customer location analysis.

Recommend improvements in product offerings based on transaction analysis.